

System Architecture Exploration and Analysis for Recurrent Neural Networks with Block-Circulant Matrices

Zhi-Ying Zhu, Pin-Yi Chou, Wei-Kai Cheng

Department of Information and Computer Engineering

Chung Yuan Christian University

Taoyuan City, Taiwan

g11277002@cycu.edu.tw, g11477007@cycu.edu.tw, wkcheng@cycu.edu.tw

摘要

本研究提出一種結合區塊循環矩陣與量子行為粒子群最佳化 (QPSO) 的遞迴神經網路架構，以解決傳統 RNN 在運算效率與超參數調整上的限制。實驗結果顯示，QPSO-LSTM 在精度指標上表現最佳，而 QPSO-BLKLSTM 在訓練時間上具顯著優勢，證明本研究方法能在精度與效率之間取得良好平衡，並展現實務應用潛力。

關鍵詞： 區塊循環矩陣、遞迴循環神經網路、長短期記憶網路、門控循環單元、超參數最佳化、量子行為粒子群最佳化

Abstract

This study proposes a recurrent neural network framework that integrates block-circulant matrices with quantum-behaved particle swarm optimization (QPSO) to address the limitations of traditional RNNs in computational efficiency and hyperparameter tuning. Experimental results show that QPSO-LSTM achieves the best accuracy, while QPSO-BLKLSTM significantly reduces training time. The findings demonstrate that the proposed approach achieves a favorable balance between accuracy and efficiency, highlighting its potential for practical applications.

Keywords: block circulant matrix, recurrent neural network, LSTM, GRU, hyperparameter optimization, quantum-behaved particle swarm optimization.

1. Introduction

Recurrent neural networks (RNNs) demonstrate

strong performance in natural language processing and time-series analysis. Variants such as LSTM and GRU alleviate the vanishing gradient problem via memory cells and gating mechanisms, thereby improving stability and predictive accuracy. However, their high computational complexity and large parameter sizes limit deployment on edge devices. To address this, we introduce block-circulant matrices for weight compression [1] and integrate quantum-behaved particle swarm optimization (QPSO) [2] to propose an automated hyperparameter search framework, aiming to improve both accuracy and efficiency.

The main contributions of this study are as follows:

1. QPSO-BLKRNN architecture: Incorporating block-circulant matrices into LSTM and GRU significantly compresses weight matrices and accelerates computation, while QPSO automatically optimizes hyperparameters, avoiding the inefficiency of manual tuning.
2. Automated search process: QPSO optimizes hyperparameters including time window size, hidden dimensions, batch size, block size, and model type, improving global exploration and reducing the risk of local optima.
3. Validated performance: Experiments on transformer oil temperature and Nanjing shared-bicycle datasets demonstrate that QPSO-BLKLSTM and QPSO-BLKGRU achieve

substantial training time reduction with minimal accuracy loss, confirming the framework's potential in edge computing and practical applications.

2. Related Research

2.1. Block Recurrent Neural Network

In order to reduce the huge number of parameters and computational burden of recurrent neural networks, researchers have proposed structured compression techniques, among which the block circulant matrix is a representative method. [1]. Its core concept is to divide the large weight matrix into several small blocks and make each block. The block conforms to the circulant matrix structure, and only the first row or column needs to be stored to fully restore the entire matrix.

In recent years, there have been many studies on block recurrent neural networks: Research [3] proposed a method that combines Block recurrence compression technology and ReRAM-based processing-in-memory (PIM) architecture. This research system uses systolic data flow and ReRAM crossbar switch array to effectively support the computational characteristics of block recurrence RNN. At the same time, this research converts the matrix multiplication operation in the fully connected layer into a series of Fourier transform (FFT) and point-by-point multiplication operations, further reducing the space and time requirements of the model. Time complexity.

In addition, studies [4][5][6] have also shown that block recurrent neural networks are more efficient than traditional LSTM. The weight matrix architecture of the recurrent neural network can effectively reduce the storage space and overall computational cost of the model. This study proposes a hybrid architecture based on block recurrent neural networks to improve the limitations of traditional LSTM in edge computing system.

2.2. Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO), proposed by Kennedy and Eberhart in 1995 [7], is inspired by natural swarm behavior, where particles iteratively adjust their positions and velocities based on individual best (pbest) and global best (gbest) solutions. In recent years, PSO has been widely applied to parameter tuning and structural design of deep learning models. For instance, study [8] combined LSTM with PSO for hyperparameter optimization, improving the accuracy and efficiency of time-series prediction, while study [9] introduced a CNN-LSTM-AP model optimized by PSO for enhanced fault prediction. These works demonstrate that PSO can effectively reduce manual tuning effort and improve model performance. To overcome local minima limitations, several variants have been proposed, including Quantum-behaved PSO (QPSO) [2], which enhances global search using quantum principles, and Multi-objective PSO (MOPSO) [10], designed for problems involving multiple optimization criteria.

2.3. Quantum Behavioral Particle Swarm Optimization (QPSO)

To overcome the shortcomings of traditional PSO, scholars proposed quantum-behaved PSO (QPSO) [2]. Different from relying on velocity vector updates, QPSO introduces quantum mechanics principles and updates particles by sampling from probability distribution. This design not only enhances global search capabilities but also enables faster convergence.

According to studies [11], [12], QPSO outperforms PSO when handling sudden peaks that are often overlooked in PSO. These peaks tend to be smoothed out, thereby reducing prediction accuracy. In contrast, QPSO is more effective in capturing abrupt fluctuations and extreme values. Motivated by this, the present study proposes integrating QPSO with block-circulant recurrent neural networks.

This improves the model's ability to perceive sudden

changes in time series, further enhances the prediction accuracy, and This algorithm is used to find the global optimal solution in a faster and more efficient way.

3. Research Methods

This section presents the proposed QPSO-Block Recurrent Neural Network architecture and optimization process. The design integrates structured compression with algorithmic optimization, enabling time-series models to operate effectively under limited computing resources while maintaining both efficiency and accuracy. The overall framework consists of three main components: block-circulant RNN construction, QPSO-based hyperparameter optimization, and bidirectional architecture integration.

3.1. Block Circulant LSTM/GRU

This study first structures the weight matrices of standard LSTM and GRU into block-circulant matrices. Taking block size k as an example, only the first row of each block needs to be stored to reconstruct the complete matrix, and the number of parameters can be reduced by about k times.

First, we customize a standard LSTM Cell / GRU Cell and transform its weight matrix into a block-circulant matrix. If the specified block size does not exactly divide the combined input dimension ($\text{input_dim} + \text{hidden_dim}$), a padding strategy is applied to extend the input vector of each time step to the nearest divisible length, thereby avoiding dimensional inconsistencies in subsequent computations.

Then, put the defined LSTM Cell / GRU Cell into the Keras layer to construct the framework of a deep learning model. By employing the Sequential model, multiple layers of block-circulant RNN structures can be stacked in series, enabling end-to-end training and prediction.

3.2. QPSO Algorithm

The second part is the hyperparameter search module based on QPSO. Different from traditional PSO that

updates the position through the velocity vector, QPSO using quantum probability distribution to update samples gives particles a higher global exploration capability.

The overall procedure for the architecture search is shown in figure 1:

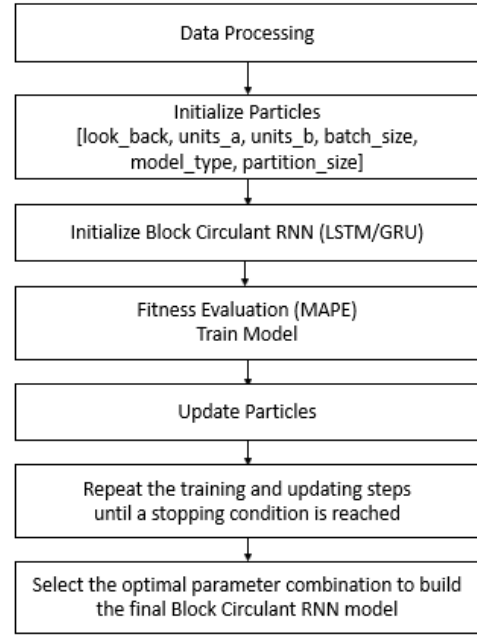


Figure1. The QPSO-BLKRNN architecture search procedure

Step 1: Preprocess each dataset, first load the data, then use MinMaxScaler.

Feature normalization is applied to scale all features into the range $[0,1]$ to avoid the influence of magnitude differences between columns. The model training stability is checked, and finally the data is split into 80% training set and 20% test set.

Step 2: Initialize a set of particle combinations, where the parameter combination includes 6 parameter particles:

- Time window size
- Number of first hidden layers
- Number of second hidden layers
- Batch size
- Block size
- Model Type

And at the same time set the maximum and minimum range of particle combination.

Step 3: Initialize the block circulatory recurrent neural network.

The block circulatory recurrent neural network in this study is composed of multiple layers of Block Circulant LSTM cell / Block Circulant GRU cell. Some parameters are crucial, for example:

- Time window size: determines the length of history that the model considers.
- Number of units in each hidden layer is the number of units in each LSTM/GRU layer.
- Block size is the smallest unit that constitutes the block circulant matrix. Its size determines the size of the entire block circulant matrix.

Step 4: Use MAPE to evaluate the fitness.

MAPE (mean absolute percent error) is the absolute percentage error between the actual value and the predicted value. The MAPE equation is as below:

$$MAPE = \sum_i^n (|y_i^{true} - y_i^{pred}| / y_i^{true}) / n * 100\% \quad (1)$$

Where y_i^{true} is the actual value, y_i^{pred} is the predicted value, and n is the number of samples. Therefore, the smaller the MAPE value, the more accurate the model is.

Step 5: Update the particle positions using the QPSO algorithm.

First, the average of all the particles' individual best positions is calculated. Then, a local attraction point is determined for each particle as a weighted combination of the individual best position and the global best position. The particle positions are then updated according to the following formula.

Step 6: Repeat the training and updating steps until a stopping condition is reached, such as the maximum number of iterations or the maximum number of convergence to optimal fitness.

Step 7: Select the optimal parameter combination to build the final block recurrent RNN model and perform testing or actual prediction.

3.3. Bidirectional Block Circulant LSTM / Block Circulant GRU

Finally, in the model testing phase, the final block recurrent neural network is composed of a two-layer bidirectional. It is constructed by Block Circulant LSTM / Block Circulant GRU. The central idea is to build two RNNs simultaneously: one processes the input sequence in the forward direction, while the other processes the reversed sequence in the backward direction. Finally, at each time step, the hidden layer states in both directions are combined as output to capture the "past" and "future" in the context.

The information is shown in figure 2:

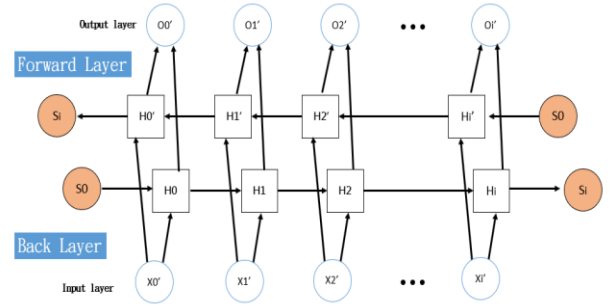


Figure 2. Bidirectional recurrent neural network

This architecture has a strong ability to understand context and can consider both past and future information. It is particularly suitable for scenarios such as language models and time series prediction, effectively improving the accuracy and stability of predictions.

4. Experimental Results

In order to verify the prediction effect of the proposed QPSO-Block Circulant RNN model, we used the Nanjing shared bicycle usage data set [13], the data set is used to predict the number of bicycles available at each station in Nanjing during peak and off-peak hours. The total number of data sets is about 648 records with timestamps, station information, and available bicycle counts during peak and off-peak hours.

The dataset was divided into training and test sets in a ratio of 8:2, and a two-layer bidirectional RNN was employed for the final prediction and analysis. Table 1.

shows the parameters recommended by PSO for each model, Table 2. shows the parameters recommended by QPSO for each model.

Table 1. the parameters recommended by PSO for each model

	PSO- BLKL STM	PSO- LSTM	PSO- BLK GRU	PSO- GRU
Suggest paramete rs	[4, 256, 128,11, 0, 64]	4, 256, 128,11, 0, 64]	[4, 256, 128, 4, 1, 61]	[4, 256, 128, 4, 1, 61]
Training time	146.191 0 sec	746.81 33 sec	426.42 07 sec	532.24 80 sec
Accuracy	82.48%	86.78%	86.88%	83.55%
RMSE	424.69	364.09	338.82	399.27
MAE	320.22	259.29	251.08	210.50
R^2	0.5786	0.6462	0.6936	0.7778

Table 2. the parameters recommended by QPSO for each model

	QPSO- BLK LSTM	QPSO- LSTM	QPSO- BLK GRU	QPSO- GRU
Suggest paramete rs	[4, 257, 128,12, 0, 64]	[4, 257, 128,12, 0, 64]	[3, 256, 128, 3, 1, 64]	[3, 256, 128, 3, 1, 64]
Training time	130.29 10 sec	512.00 20 sec	448.33 71 sec	771.88 86 sec
Accuracy	84.51%	85.4%	81.66%	84.06%
RMSE	384.22	375.34	480.19	400.04
MAE	188.77	291.39	361.86	300.16
R^2	0.6060	0.6240	0.3853	0.5734

As shown in Table 1, PSO-LSTM achieves a high predictive accuracy of 86.78%, but its training time is the longest among the four models. In contrast, PSO-BLKGRU attains an accuracy of 86.88% with an R^2 of 0.6936, and records the lowest error values across all evaluation metrics, while requiring only 426.43 seconds for training. These results suggest that PSO-

BLKGRU provides an effective balance between accuracy and efficiency. The PSO-GRU achieves an accuracy of 83.55% and an R^2 of 0.7778, with a training time of 532.24 seconds, showing moderate predictive ability and computational cost. Finally, PSO-BLKLSTM demonstrates excellent time efficiency, and although its accuracy is slightly reduced compared to other models, it still maintains a reasonable level of predictive performance.

The QPSO-LSTM achieved the best accuracy of 85.4% and highest R^2 of 0.624, confirming its strong predictive capability, though with longer training time of 512 seconds. In contrast, QPSO-BLKLSTM significantly reduced training time to 130 seconds while maintaining reasonable accuracy, showing its advantage in efficiency. For GRU models, QPSO-GRU delivered higher accuracy of 84.06% and explanatory power than QPSO-BLKGRU, but at the cost of substantially longer training time, whereas QPSO-BLKGRU offered faster computation with some loss in accuracy.

5. Conclusions

This study proposed a recurrent neural network architecture that integrates block-circulant matrices with the Quantum-behaved Particle Swarm Optimization (QPSO) algorithm. Experimental results show that QPSO-LSTM achieves the best accuracy, while QPSO-BLKLSTM offers the highest training efficiency; for GRU models, the conventional GRU attains better accuracy, whereas the block-circulant GRU achieves greater efficiency. Overall, the proposed framework balances accuracy and efficiency, demonstrating strong potential for practical applications. Future research may further investigate the performance of this architecture on larger-scale datasets and diverse time-series forecasting tasks, as well as explore its implementation on FPGA hardware platforms to validate its applicability in edge computing and real-time prediction scenarios.

6. References

- [1] Z. Li, C. Ding, J. Xie, H. Zhang, B. Yuan, Y. Wang, and Y. Liang, “E-RNN: Design optimization for efficient recurrent neural networks in FPGAs,” in *Proc. IEEE Int. Symp. High Performance Computer Architecture (HPCA)*, Washington, DC, USA, 2019, pp. 69–80, doi: 10.1109/HPCA.2019.00028.
- [2] J. Sun, B. Feng, and W. Xu, “Particle swarm optimization with particles having quantum behavior,” in *Proc. Congress on Evolutionary Computation (CEC)*, Portland, OR, USA, 2004, vol. 1, pp. 325–331, doi: 10.1109/CEC.2004.1330875.
- [3] N. Challapalle, S. K. Venkataramani, S. Venkataramani, K. Roy, and A. Raghunathan, “PSB-RNN: A processing-in-memory systolic array architecture using block circulant matrices for recurrent neural networks,” in *Proc. Design, Automation & Test in Europe Conf. & Exhibition (DATE)*, Grenoble, France, 2020, pp. 180–185, doi: 10.23919/DATE48585.2020.9116469.
- [4] S. Wang, Z. Li, C. Ding, B. Yuan, Q. Qiu, Y. Wang, and Y. Liang, “C-LSTM: Enabling efficient LSTM using structured compression techniques on FPGAs,” in *Proc. ACM/SIGDA Int. Symp. Field-Programmable Gate Arrays (FPGA)*, 2018, pp. 11–20, doi: 10.1145/3174243.3174253.
- [5] C. Ding, S. Liao, Y. Wang, Z. Li, N. Liu, Y. Zhuo, C. Wang, X. Qian, Y. Bai, G. Yuan, X. Ma, Y. Zhang, J. Tang, Q. Qiu, X. Lin, and B. Yuan, “CirCNN: Accelerating and compressing deep neural networks using block-circulant weight matrices,” *arXiv preprint arXiv:1708.08917*, 2017.
- [6] M. A. Alhartomi, A. A. A. El-Moursy, A. G. Radwan, and A. M. Eltawil, “Low-area and low-power VLSI architectures for long short-term memory networks,” *IEEE J. Emerg. Sel. Topics Circuits Syst.*, vol. 13, no. 4, pp. 1000–1014, Dec. 2023, doi: 10.1109/JETCAS.2023.3330428.
- [7] J. Kennedy and R. Eberhart, “Particle swarm optimization,” in *Proc. Int. Conf. Neural Networks (ICNN)*, Perth, WA, Australia, 1995, vol. 4, pp. 1942–1948, doi: 10.1109/ICNN.1995.488968.
- [8] Y. Xu, C. Hu, Q. Wu, S. Jian, Z. Li, Y. Chen, G. Zhang, Z. Zhang, and S. Wang, “Research on particle swarm optimization in LSTM neural networks for rainfall-runoff simulation,” *J. Hydrology*, vol. 608, p. 127553, 2022, doi: 10.1016/j.jhydrol.2022.127553.
- [9] Y. Zhang, P. Zhang, W. Zhang, and M. Wang, “CNN-LSTM-attention with PSO optimization for temperature and fault prediction in meat grinder motors,” *Discover Appl. Sci.*, vol. 7, 2025, doi: 10.1007/s42452-025-07011-3.
- [10] M. Al-Behadili, D. Ouelhadj, and D. Jones, “Multi-objective particle swarm optimization for robust dynamic scheduling in a permutation flow shop,” in *Advances in Intelligent Systems and Computing*, vol. 557, Springer, 2017, pp. 498–507, doi: 10.1007/978-3-319-53480-0_49.
- [11] W. Li, “Load forecasting method based on QPSO-LSTM,” *Appl. Comput. Eng.*, vol. 142, pp. 18–22, 2025, doi: 10.54254/2755-2721/2025.KL21491.
- [12] S. Khan, T. Mazhar, T. Shahzad, T. Ali, M. Ayaz, Y. Ghadi, E.-H. Aggoune, and H. Hamam, “Optimizing load demand forecasting in educational buildings using quantum-inspired particle swarm optimization (QPSO) with recurrent neural networks (RNNs): A seasonal approach,” *Sci. Rep.*, vol. 15, p. 19349, 2025, doi: 10.1038/s41598-025-04301-z.
- [13] M. Cao, Y. Liang, Y. Zhu, G. Lü, and Z. Ma, “Prediction for origin-destination distribution of dockless shared bicycles: A case study in Nanjing City,” *Front. Public Health*, vol. 10, p. 849766, 2022, doi: 10.3389/fpubh.2022.849766.